MILITARY SPECIFICATION

BEACON SET, RADAR

AN/APN-202

This Specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall receive interrogations from X-band radars and transmit replies to appropriate interrogations at a different frequency within the X-band.

1.2 Classification - The equipment covered by this specification shall consist of the following items:

	Type Designation	Applicable Paragraph
Receiver-Transmitter Radar	RT-1028/APN-202	3.5.1
Control, Radar Beacon	C-8752/APN-202	3.5.2
Antenna	AS-2619/APN	3.5.3

1.3 Associated Equipment - This equipment shall operate with the associated equipment listed in 6.8.

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

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SPECIFICATIONS

Military

MIL-W-5088	Wiring; Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-C-6781	Control Panel; Aircraft Equipment, Rack or Console Mounted
MIL-P-7788	Panels, Information, Integrally Illuminated
MIL-M-7793	Meter, Time Totalizing
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures; Preproduction and Acceptance for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-R-81614	Receiver, Radar Augmentor R-1623/APN; Modulator, X-Band Beacon MD-817/APN and Antenna, Ka-Band Radar AS-2406/APR

STANDARDS

Military

MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-785	Reliability Program for System and Equipment Development and Production

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MIL-STD-794 Parts and Equipment, Procedures for Packaging and Packing of

Naval Air Systems Command

AR-5 Microelectronic Devices Used in Avionics Equipment, Procedures for Selection and Approval of

AR-10 Maintainability of Avionics Equipment and Systems, General Requirements for

AR-34 Failure Classification for Reliability Testing, General Requirements for

ET-564 Installation Instructions for Radar Beacon AN/APN-202

2.1.1 Availability of Documents

(1) When requesting specifications, standards, drawings, and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Publication and Forms Center, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

3. REQUIREMENTS

3.1 Preproduction - This specification makes provision for preproduction testing.

3.2 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

(1) Microelectronic technology shall be considered and microelectronic items shall conform to requirements specified herein.

(2) Other parts and materials requirements shall conform to Specification MIL-E-5400.

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(3) Non-repairable subassemblies shall be used in accordance with Specification AR-10 and as outlined in Specification MIL-E-5400.

(4) When previously produced models of this equipment did not use non-repairable subassemblies, the design shall not be changed to employ non-repairable assemblies without the approval of the procuring activity.

3.2.1 Non-Standard Parts and Material Approval - Approval for the use of non-standard parts and materials (including electron tubes, transistors and diodes) other than microelectronic devices shall be obtained as outlined in specification MIL-E-5400. Microelectronic devices shall be approved as outlined in AR-5.

3.2.2 Microelectronic Modular Assemblies - When used, microelectronic Modular Assemblies shall meet the requirements of AR-5.

3.2.3 Modules - The electronics portions of the equipment shall be functionally modularized in accordance with AR-10.

3.3 Design and Construction - The equipment shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein.

3.3.1 Total Weight - The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed the values listed below:

Receiver-Transmitter, Radar	5.5 pounds
Control, Radar Beacon	1.0 pounds
Antenna	1.4 pounds

3.3.2 Reliability - The contractor shall conduct a reliability program using MIL-STD-785 as a guide. On reorder from a supplier who has previously produced the equipment, the program previously used may be continued unless otherwise indicated in the contract or order.

3.3.2.1 Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently for a period of at least 150 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

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3.3.2.2 Operating Life - The equipment shall have a total operating life of 6000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

3.3.2.3 Reliability in Mean Time Between Failure (MTBF) - The equipment shall have 1000 hours of mean (operating) time between failures when tested and accepted as outlined under the requirements of 4.4.2.2.1. Normal scheduled maintenance for the magnetron is permissible during these tests if required. Mean operating life of the magnetron shall be at least 400 hours.

3.3.2.4 Time Totalizing Meter - The following units shall contain time totalizing meters in accordance with Specification MIL-M-7793.

Unit	Type of Meter
RT-1028/APN-202	MS-17321-9

3.3.3 Cabling and Connections

3.3.3.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400.

3.3.3.2 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

3.3.4 Control Panels - All rack or console mounted control panels shall conform to the applicable requirements of Specification MIL-C-6781. The configuration of all panels must be approved by the procuring activity prior to preproduction testing.

3.3.5 Interchangeability - The equipment shall meet the interchangeability requirements of Specification MIL-E-5400.

3.3.6 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall meet the requirements of MIL-STD-461.

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3.3.7 Test Signals - The equipment shall produce, at the output of each significant subassembly, a signal indicative of the operating condition of that module. These signals shall be in form of a logic "0" for NO-GO and a logic "1" for a GO condition. Fault location to module level shall easily be accomplished with these signals. (See 3.5.1.26.1)

3.3.8 Nomenclature and Nameplates - Nomenclature and serial number assignment, nameplate approval and equipment identification shall be in accordance with Specification MIL-N-18307.

3.3.9 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room Ambient (25°C ± 5°C)
Altitude	Normal Ground
Vibration	None
Humidity	Room ambient up to 90 percent relative humidity
Input Power Voltage	27.5 ± 0.5 VDC

3.3.10 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class 2 equipment, except as modified herein:

3.3.10.1 Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements of Curve IV of Specification MIL-E-5400, except that the console control panel shall meet Curve I from 5 Hz to 52 Hz and Curve III from 52 Hz to 500 Hz.

3.3.10.2 Temperature - The equipment shall operate satisfactorily under the temperature versus altitude conditions of MIL-E-5400 for continuous operation except that column II of Table I shall not apply, and column VII shall be limited to the temperatures listed in column I. The transponder must be mounted to an adequate heat sink. Maximum power input to the equipment will be 45 watts. Under this condition, the heat sink shall not exceed 72°C.

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3.3.11 Warm-Up Time - The time required for the equipment to warm-up prior to operation shall be kept to a minimum and shall not exceed 5 minutes under standard conditions and 15 minutes at extreme service conditions.

3.3.12 Input Electrical Power

3.3.12.1 Operating Power - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance when energized from the following power sources having characteristics and limits as defined in MIL-STD-704. The power required shall not exceed the specified amounts.

(1) DC Power, 28V, Category B, 1.5 amps.

3.3.12.2 Lighting Power - Input power for lighting shall require not more than 0.5 amp at 5 volts.

3.3.12.3 Degraded Performance - Degraded performance will be permitted for voltage transients not exceeding 0.5 second during normal electrical system operation. Operation shall return to normal with no resulting damage to the equipment.

3.3.13 Cooling - The equipment shall operate satisfactorily under the specific thermal characteristics described by Curve A of Figure 3 of MIL-E-5400 for Class 2 equipment. No cooling air shall be required. Refer to 3.3.10.2

3.4 Performance - Unless otherwise specified, values set forth to establish the requirements of satisfactory performance apply to performance under both standard and extreme service and input power conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.4.1 Operation - The equipment shall receive and decode interrogations and generate single or double pulse replies to these interrogations from X-band radars. The Radar Beacon Set shall have a common antenna terminal for both receiver and transmitter functions. Processed video signals derived from the radar interrogations and transponder replies, shall be provided for application to external equipments. In addition, the transmitter be triggered and its output amplitude modulated by external equipments such as the ACLS radar augmentor receiver R-1623/APN. Functions performed by the radar beacon and by the ACLS radar augmentor receiver shall be controlled and switchable at the control panel. The control shall contain circuitry for initiating and indicating the results of the Built-In-Test (BIT) function in the radar beacon and augmentor receiver.

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3.5 Detail Requirements

3.5.1 Receiver-Transmitter, Radar RT-1028/APN-202 - The receiver-transmitter shall meet the following requirements:

3.5.1.1 Receiving and Transmitting - The receiver-transmitter shall operate from a single antenna for receiving and transmitting. The transmitter and receiver frequencies shall always be separated by at least 50 MHz.

3.5.1.2 Impedance - The nominal input and output impedance of the transponder shall be 50 ohms.

3.5.1.3 Delay

3.5.1.3.1 Fixed Delay - The receiver-transmitter shall have one (1) delay setting. This shall be 1.0 ±0.1 usec at an input signal level of 0 dBm.

3.5.1.3.2 Delay Variation with Signal Level - The total (absolute) variation in delay shall not exceed .05 usec for signal levels between 0 dBm and -62 dBm.

3.5.1.3.3 Delay Jitter - Delay Jitter, which is defined as pulse to pulse variation in delay at a constant input signal level, shall not exceed the following values:

(a) For signal levels of 0 dBm to -55 dBm:
0.02 usec peak to peak

(b) For signal levels of -55 dBm to -62 dBm:
0.05 usec peak to peak

3.5.1.4 Pulse Recurrence Frequency (PRF) - The equipment shall be capable of operation at any prf from 10 to 4000 pps, unless this would result in exceeding the duty cycle limits specified, in which case the maximum prf shall be that corresponding to the maximum allowable duty cycle.

3.5.1.5 Random Triggering - The equipment shall not generate random replies (those replies not synchronized with interrogations) at a rate greater than 10 pps under any operation conditions specified herein.

3.5.1.6 Receiver Frequency - The center frequency of the receiver-transmitter's receiver passband shall be adjustable to any frequency from 8500 to 9500 MHz. Center frequency is defined as the mid-point frequency between the receiver passband 3 dB points.

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3.5.1.7 Receiver Frequency Stability - Under any operating conditions listed herein, except during changes in ambient temperature, the receiver frequency shall not vary more than ±3 MHz after warm-up as specified in paragraph 3.3.11. During changes in ambient temperature, the co-efficent of frequency drift shall not exceed 75 kHz per degree C. However, changes in receiver frequency due to all combinations of operating conditions including temperature shall not require a change in the interrogating signal frequency in order to meet the sensitivity requirements of 3.5.1.8.

3.5.1.8 Sensitivity and Dynamic Range - The receiver-transmitter shall reply to at least 99 percent of the interrogation signals at any input level between 0 dBm and -65 dBm. In the coded mode of operation, the sensitivity shall be measured at the optimum code spacing, which shall be within ±0.1 usec of the nominal values. The receiver shall be capable of receiving, without damage, signals up to +20 dBm.

3.5.1.9 Bandwidth - The receiver bandwidth shall not be less than 10 MHz nor more than the 20 MHz as measured at the points corresponding to a 3 dB decrease from the maximum sensitivity.

3.5.1.10 Image - The image frequency shall be at least 100 MHz from the receiver frequency. At the image frequencies, the receiver shall exhibit 60 dB decrease in sensitivity compared to that at the frequency to which the receiver is tuned.

3.5.1.11 Input Signal - The receiver-transmitter shall be capable of operation from several types of input signals depending on the code chosen. A choice of single-pulse operation with any one of five double-pulse codes shall be possible when used with the external (remote) control described herein.

3.5.1.11.1 Single Pulse Operation - In this mode of operation, the transmitter shall reply to a single pulse of the following characteristics:

(a) Pulse Width	0.3 to 3 usec
(b) Rise Time	0.1 usec or less
(c) Fall Time	0.2 usec or less

3.5.1.11.2 Double-Pulse Operation - The receiver-transmitter shall respond to any one of five spacings of double-pulse interrogation as selected by the control panel, paragraph 3.5.2.. The standard double-pulse spacings are as follows:

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(a) 1.9 ± 0.1 usec	CODE 1
(b) 2.8 ± 0.1 usec	CODE 2
(c) 3.7 ± 0.1 usec	CODE 3
(d) 4.6 ± 0.1 usec	CODE 4
(e) 5.5 ± 0.1 usec	CODE 5

Individual pulse characteristics are:

(a) Pulse Width	0.3 to 0.75 usec
(b) Pulse Rise Time	0.1 microsecond, or less
(c) Pulse Fall Time	0.2 microsecond, or less
(d) Frequency	Within ± 2 MHz of the receiver center frequency

3.5.1.11.3 Double Pulse Rejection Characteristics - The receiver-transmitter shall not respond to coded interrogations that are ±0.7 usec or greater from the nominal spacing listed in 3.5.1.11.2.

3.5.1.12 Transmitter - The transmitter frequency shall be adjustable to any frequency from 8800 to 9500 MHz.

3.5.1.13 Transmitter Frequency Stability - Under any operating conditions listed herein, except during changes in ambient temperature, the transmitter shall not drift more than ±3 MHz, and the drift rate shall not exceed 1 MHz per minute after the warm-up specified in paragraph 3.3.11. During changes in ambient temperature, the temperature co-efficient of frequency drift of the transmitter shall not exceed 50 kHz per degree C. This shall include any frequency modulation present during service conditions listed herein.

3.5.1.14 Power Output - The peak power output of the transmitter shall be 300 watts minimum under all operating conditions listed herein. Power output shall be measured at the receiver-transmitter antenna connector, J1.

3.5.1.15 Transmitter Duty Cycle - The transmitter shall meet the requirements of this Specification at any duty cycle up to .002.

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3.5.1.16 Overinterrogation Protection - The receiver-transmitter shall not be damaged when interrogated at prf's which would result in exceeding the cuty cycle limitation. Within 1,000 usec, after the interrogation prf falls to the maximum specified value, or less, the transponder shall meet all requirements of this document.

3.5.1.17 Transmitter Pulse Characteristics - The transmitter output pulse shall have the following characteristics:

(a) Pulse Width: 0.3 ± 0.1 usec

(b) Pulse Width Jitter: Not to exceed 0.01 usec peak-to-peak at the one-half power points.

(c) Rise Time: Not to exceed 0.1 usec from 10 percent to 90 percent amplitude points.

(d) Fall Time: Not to exceed 0.2 usec from 90 percent to 10 percent amplitude points.

(e) Spectrum: The rf bandwidth (in megacycles) of the transmitter pulse, measured at one-quarter power by means of an rf spectrometer, shall not exceed 3.0 divided by the pulse width in microseconds.

3.5.1.18 Recovery Time - When operated in either the single pulse or coded interrogation mode within the duty cycle limitations, the transmitter shall be capable of replying (with single-pulse replies) to each of three (3) interrogations (code groups) when the spacing between any two (2) successive interrogations is a minimum of 6 usec. These requirements shall be satisfied even though the signal level of adjacent interrogations differs by as much as 60 dB. When operated in the external data trigger mode, the transponder shall be capable of replying with up to five consecutive pulses spaced 6 usec apart.

3.5.1.19 Coded Reply Pulse Operation - The receiver-transmitter shall provide either single or double pulse (encoder) reply operation as selected by the control panel, paragraph 3.5.2. The standard double-pulse spacings shall be as follows:

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Code	Spacing
S	Single Pulse
1	49 ± 4.9 usec
2	62 ± 6.2 usec
3	99 ± 9.9 usec
4	124 ± 12.4 usec
5	148 ± 14.8 usec

3.5.1.20 External Trigger Input - Two inputs shall be provided for external triggering at the transmitter. A positive pulse having an amplitude from 7 to 11 volts, measured across 100 ohms, shall be required to trigger the transmitter.

3.5.1.21 Detected RF Output - A video pulse whose leading edge is indicative of the reply transmission time shall be provided. Amplitude of this pulse shall be -0.5 volts minimum when measured across 100 ohms.

3.5.1.22 External Modulation - The transponder shall contain a modulator capable of applying amplitude modulation to the transmitter reply pulse. The modulator shall be capable of producing a 3 dB variation in peak power output and be switchable in 1 usec or less. A transponder front-panel jack shall be provided to accept the external modulating signal. Nominal modulating frequency is 100 Hz.

3.5.1.23 Video Output - A processed receiver video output shall be provided. Video signals shall be 9 ± 2 volt positive pulses when measured across a 100 ohm load.

3.5.1.24 Form Factor - The receiver-transmitter shall be enclosed in a case not to exceed 6.88 x 2.82 x 6.8 inches excluding connectors. Reference EI-564.

3.5.1.24.1 Weight - The weight of the receiver-transmitter shall not exceed 5.5 lbs.

3.5.1.25. Controls - The following controls shall be provided on the external surface of the receiver-transmitter:

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(a) Local Oscillator Tuning

(b) Transmitter Tuning

(c) Pre-selector Tuning

3.5.1.26 Test Points

3.5.1.26.1 External Test Points - The following signals shall be made available at the receiver-transmitter power connector for use in alignment and failure diagnosis:

(a) Receiver Video - Unprocessed video from the i-f amplifier.

(b) Test Signals - A logic level signal shall be supplied that is indicative of the operational status of the replaceable modules. The module test output, logic level requirements listed below are measured under standard conditions. The voltages are open circuit measurements that apply for interrogation rates of 800 pps minimum. Output impedances shall not exceed 5600 ohms. Accidental shorting of these test outputs to ground shall not result in permanent damage to the circuits.

TEST OUTPUT	LOGIC "0"	LOGIC "1"
Power Supply	+0.75 volts max.	+1.5 volts min.
Receiver Video	+0.75 volts max.	+1.5 volts min.
Modulator	+0.5 volts max.	+1.0 volts min.
Pulse Assembly	+0.75 volts max.	+1.5 volts min.

3.5.1.26.2 Internal Test Points - Sufficient test points shall be provided internally to allow rapid isolation of malfunctions to a particular section of the receiver-transmitter.

3.5.1.27 External Connectors - The External Connections to the receiver transmitter shall consist of the following:

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Reference Designation	Connector Part No.	Function
J1	GRFF 4075-0007	Antenna Connection
J2	MS3114H14-19P	Power Control and Test
J3	GRFF 4068-0001	External Data Trigger
J4	GRFF 4068-0001	External Trigger
J5	GRFF 4068-0001	Video Output
J6	GRFF 4068-0001	Detected RF
J7	GRFF 4068-0001	External Amplitude Modulation

3.5.1.28 Content - The receiver-transmitter shall contain the following sub-assemblies:

1. RF-Assembly
2. IF-Amplifier
3. Decoder-Encoder
4. Modulator
5. Power Supply

3.5.2 Control, Radar Beacon C-8752/APN-202 - The control shall meet the applicable requirements of Specification MIL-C-6781 for Type I Control Panel. The lighting of the plastic panel shall conform to MIL-P-7788 Class 1-R, Type IV with a lamp voltage of 5 volts.

3.5.2.1 Function - The control shall provide a means of setting the operational mode and selecting the desired interrogation and reply code structures.

3.5.2.2 Controls - The control panel shall have the following controls:

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(a) "MODE" - A six-position rotary switch for operational mode selection. Positions shall be as follows:

OFF - Power removed from all equipment

STBY - (Standby) Power is applied to the receiver-transmitter, but the transmitter shall not respond to interrogations.

BCN - (Beacon) The receiver-transmitter is fully operative and shall respond to valid X-band interrogations.

TEST - A BIT signal is applied to the receiver-transmitter and the ACLS Augmentor receiver. The test signal shall cause the transmitter to generate X-band reply signals. The X-band replies shall be detected and cause a "GO" indication at the control panel test lamps. Separate lamps shall be provided for the receiver-transmitter and augmentor receiver.

ACLS - Power is applied to the ACLS augmentor receiver and the receiver-transmitter receiver is placed in standby. The receiver-transmitter's transmitter shall be triggered by signals from the augmentor receiver.

XDAT - (External Data) The receiver-transmitter's decoder and encoder circuits shall be disabled, and these functions performed by external equipments. Receiver video shall be available for processing at the video output terminal (1J5), and replies shall be initiated by external triggers to the modulator at 1J3.

(b) "DECODE" - A six position rotary switch for selecting the receiver-transmitter address code shall have the following positions:

S - Single Pulse

1 - Double Pulse Code 1

2 - Double Pulse Code 2

3 - Double Pulse Code 3

4 - Double Pulse Code 4

5 - Double Pulse Code 5

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(c) "ENCODE" - A six-position rotary switch for selecting the receiver-transmitter identification reply code shall have the following positions:

S - Single Pulse

1 - Double Pulse Code 1

2 - Double Pulse Code 2

3 - Double Pulse Code 3

4 - Double Pulse Code 4

5 - Double Pulse Code 5

3.5.2.3 Form Factor - The control dimensions shall not exceed 1.50 x 3.57 x 5.75 inches including connectors. Reference EI-564.

3.5.2.4 Weight - The weight of the control shall not exceed 1 pound.

3.5.2.5 Electrical Connections - External connectors on the control panel shall be as follows:

Reference Designation	Connection Part No.	Function
J1	MS3112E-14-19P	Transponder Power, Control and Test
J2	MS3112E-10-6S	ACLS Augmentor Receiver power and test

3.5.3 Antenna AS-2619/APN - The antenna shall meet the following requirements:

3.5.3.1 Type - Slot Array

3.5.3.2 Operating Frequency Range - 9310 MHz to 9415 MHz

3.5.3.3 Polarization - Horizontal

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3.5.3.4 Pattern Coverage - When mounted on a suitable ground plane, the antenna shall have the following coverage:

(a) Elevation - The 1/2 power beamwidth shall be 14 degrees minimum.

(b) Azimuth - Omni-directional ± 2 dB.

3.5.3.5 Input Impedance and VSWR - The nominal input impedance shall be 50 ohms with a maximum VSWR of 1.7:1 maximum.

3.5.3.6 Gain - The gain, with respect to a linear isotropic shall be plus 3 dB minimum at peak of beam and 0 dB nominal at 14 degrees above the ground plane (horizontal).

3.5.3.7 Form Factor - The form factor shall be as indicated in EI564-3.

3.5.3.8 External Connector - The external connector to the antenna shall mate with a type TNC male connector.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements.

(1) Preproduction (First Article) Tests

(2) Initial Production Tests

(3) Acceptance Tests

(4) Life Tests

4.2 Preproduction (First Article) Tests - Preproduction tests shall be conducted by the contractor on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the approved test procedure of 4.6. The Government Inspector and the procuring activity shall be advised when tests are to be conducted so that a government representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 Preproduction (First Artilce) Test Data - The contractor shall submit all data collected in conducting these tests to the procuring activity for review.

4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification, other applicable specifications and the contract. Preproduction tests shall include environmental tests in accordance with the procedures of Specification MIL-T-5422.

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4.2.3 Preproduction (First Article) Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Pre-fabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Initial Production Tests - One of the first ten (10) production equipments shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. No other tests shall be conducted on the equipment prior to starting the Initial Production Tests. The preproduction sample shall not be selected for this test.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance, and quality and that it meets all applicable requirements.

4.3.2 Accessory Material - In addition to the complete equipment submitted for Initial Production Tests the contractor shall also submit such accessory material and data necessary to test the equipment.

4.3.3 Initial Production Sample Approval - Approval of the Initial Production Sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Production Sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and

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accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.4 Reconditioning of Initial Production Test Sample - On completion of the initial production test the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government Inspector. Contractors not having adequate facilities for conducting all required tests shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptable tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Reliability Assurance Tests

(3) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Run in Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

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4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Run-in Test (MRIT) - Prior to submission of the equipment for the all equipment tests described in 4.4.2.2.1 the equipment shall be subjected to preconditioning operation in accordance with the environmental conditions of test level E of MIL-STD-781 (less vibrations) for a period of not less than 75 hours which shall include a minimum of 12 temperature cycles. An additional 10 hours of MRIT shall be conducted on each equipment under the same environmental conditions including vibration. Failure reporting as required by MIL-STD-781 shall commence only with the beginning of the additional 10 hour test period portion of MRIT.

4.4.2 Reliability Assurance Tests - Reliability Assurance Tests shall be conducted using MIL-STD-781. Tests as required by both the Qualification Phase and the Production Acceptance Phase shall be conducted. Classification of failure shall be in accordance with MIL-STD-781 and AR-34.

4.4.2.1 Qualification Phase - Prior to the acceptance of equipments under the contract or order, a minimum of three (3) equipments shall be tested as outlined in MIL-STD-781, under the section entitled, "Qualification Phase of Production Reliability Tests." The maximum number of equipments to be used shall be those listed in Table 5 of MIL-STD-781. For the Qualification Phase, Test Level "E" shall be used. The Accept-Reject Criteria for Test Plan III shall be used.

4.4.2.2 Reliability Production Acceptance Phase Tests - The production equipment shall be tested as outlined in MIL-STD-781 (as modified herein) under the Section entitled, "Production Acceptance (Sampling) Phase of Production Reliability Tests." For the All-Equipment Test, Test Level "E" shall be used.

4.4.2.2.1 All Equipment Test - Each equipment produced shall be tested for 75 hours. Prior to the 75 hour test on each equipment, a burn-in period may be used at the option of the contractor. If the burn-in period is to be used, the details thereof must be included in the approved test procedures. To determine whether the MTBF is being met at any time during the contract, the operating test hours and the failures thereon (not counting burn-in failures or burn-in operating time) shall be totaled and the results compared with the reject line of Test Plan III of MIL-STD-781. (Extend the line as necessary to accomodate the data). These totals shall accumulate so that at any one time the experience from the beginning of the contract

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is included. At the conclusion of each month, the test results shall be sent to the procuring activity and to the Naval Air Systems Command, Attn: Avionics Division. At any time that the current totals of test hours and test failures plotted on Test Plan III curves show a reject situation, the procuring activity shall be notified. The procuring activity reserves the right to stop the acceptance of equipment at any time that a reject situation exists pending a review of the contractor's efforts to improve the equipment, the equipment parts, the equipment workmanship, etc., so that the entire compilation will show other than a reject decision.

4.4.2.3 Test Details - The test details such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, preventive maintenance to be allowed during the test, etc., shall be part of the test procedures to be submitted and approved by the procuring activity prior to the beginning of the Qualification Test Phase of the Reliability Assurance Tests.

4.4.3 Special Tests - Special tests shall be conducted for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

4.4.3.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate that additional tests are required. (This will be determined by the procuring activity).

4.4.3.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.4 Equipment Failure - Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

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(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the Government Inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.6 Reconditioning of Tested Equipment - Equipment which has been subjected to initial production acceptance and life tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.7 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.8 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government Inspector.

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5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-E-17555 and MIL-STD-794. In the event the equipment is not covered in Specification MIL-E-17555, the method of preservation for Level A shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794.

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6. NOTES

6.1 Intended Use - The function of this beacon set is to extend the tracking range of X-band radar sets, having X-band beacon receivers, by receiving pulse type signals fro m the radar and transmitting pulse replies in the same frequency band. The beacon also has 2-pulse received signal decoding and transmission encoding capabilities when required for identification or to prevent interrogation by single pulse radars when operating in areas having a high number of X-band radars. In addition, by selection of the proper operating mode, the transmitter may be triggered by external pulses for use with ACLS or for transmitting data in the form of X-band pulses. For use with ACLS, provision has been made for amplitude modulating these X-band reply pulses.

6.2 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(1) Title, number, and date of this specification.

(2) Selection of applicable levels of packaging and packing (See 5.1).

6.3 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.4 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this

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specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.5 Non-Repairable Subassemblies - As a general rule, non-repairable subassemblies should be encapsulated or hermetically-sealed. The number of connections internal to the subassembly should be held to a minimum Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the non-repairable subassembly should evidence a Mean-Time-to-Failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

6.6 Type Designations - The type designation may be modified by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.3.8. The correct type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.7 Revisions - In specification revisions and superseding amendments an asterisk "*" preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notions. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the asterisk notations and relationship to the last previous issue.

6.8 Associated Equipment - The beacon set shall operate in conjunction with the following associated equipment:

6.8.1 Radar Augmentor Receiver R-1623/APN - When used in the All-Weather Carrier Landing System (ACLS) Mode, the beacon set shall operate in conjunction with the Augmentor Receiver as specified in MIL-R-81614.

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6.9 Definitions - Definitions pertaining to the fabrication of equipment as stated herein shall be as recorded in MIL-E-5400, "Electronic Equipment, Aircraft, General Specification for," with the following exceptions:

1. Rise Time: The time required for the leading edge of a voltage pulse to increase from 10 percent to 90 percent of the amplitude of the pulse.

2. Fall Time: The time required for the trailing edge of a voltage pulse to decrease from 90 percent to 10 percent of the amplitude of the pulse.

3. Pulse Width: The time interval between the 50 percent amplitude points of a pulse voltage except for detected r-f pulses for which the pulse width is defined as the time interval between the half power points (-3 dB points).

4. Delay and Spacing of Pulses: The time interval between 50 percent amplitude points on the leading edges of the voltage pulses under consideration.

5. Sensitivity: The minimum signal level, measured at the Receiver-Transmitter antenna terminal, which causes the transponder to reply to at least 99 percent of the interrogations per second.

6. Random Triggering: Transmissions by the transponder which are not synchronized with interrogations.

7. Pulse Recurrence Frequency (PRF): The number of pulses, or pulse groups (in the case of two pulse codes), which occur per second.

8. Satisfactory Performance: The term "Satisfactory Performance" shall mean the operation of the equipment within the applicable limits of this specification as delineated in the test procedures.

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9. Warmup: The elapsed time from initial application of d-c operating voltage after which the transponder shall meet the requirements of this technical exhibit.

10. Flange Temperature: The average of the mean temperature of the exterior surfaces of the mounting (feet) regardless of how the transponder is mounted.

6.10 This specification is under the cognizance of AIR-5333D21.

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Naval Air Systems Command
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-B-81876(AS) BEACON SET, RADAR AN/APN-202

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☒ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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